

Starvivors 2.0 Full-System Polish Audit

Prepared: 2026-05-24

Repo basis: local Starvivors workspace, current clean working tree, build passing with `npm.cmd run build`.

Sources reviewed: 2.0 GDD/build plan, phase notes, GameScene owned runtime fields, data registries, UI modules, system modules, asset folders, Forge registry, and source TODO/deferred markers.

Summary: The game is broad enough that polish should now be tracked as a whole-product pass, not only as isolated phase work. The strongest implemented pillars are the 2.0 run loop, mission/free-range selection, ship/loadout basics, Pulse Cannon, sector generation, squads, mothership/rare event prototypes, reward persistence, HUD/radar/results work, and the initial Asset Forge pipeline.

Main risk: Many systems are playable but not yet cohesive: menus use the new cockpit treatment unevenly, art direction is split between Forge/vector/procedural/PNG, audio is absent, accessibility/settings are placeholder-heavy, and several active objects still rely on scene-owned Phaser lifecycle code rather than focused ownership.

Status Scale

Status	Meaning
Implemented	Works in the current build and has at least one real user path. Still may need polish.
MVP	Playable or usable but rough, incomplete in presentation, balance, content depth, or edge cases.
Prototype	Framework exists and proves direction, but needs content, final presentation, tuning, and stronger verification.
WIP	Active work has begun but the product surface is not settled.
Missing/Mentioned	Mentioned in docs, data, or plans but absent, placeholder-only, or not promoted to live gameplay.
Risk	Works now but has maintenance, readability, scaling, UX, or performance risk.

Top Polish Priorities

Priority	Why It Matters	Suggested Shape
1. Whole-run clarity	Players need to understand why they launched, what to do, when they are done, and what they earned.	Close Phase 17 with mission tracker, radar, result copy, and eject/continue language reviewed in one pass.
2. Asset cohesion	The project mixes PNG ships, procedural enemies, Forge textures, generated candidates, and UI graphics.	Promote Forge replacements for all core ships, all live enemy silhouettes, mission/radar icons, and weapon/upgrade icons.
3. Audio and feedback	No audio assets were found. Dangerous actions lack sound identity.	Start Phase 18 with a small sound set: player fire, enemy fire, hit, hull damage, pickup, upgrade, low fuel, mission complete, eject, black hole warning.
4. Menu finish	Command, hangar, shop, settings, pause, upgrade, and results all exist but need final hierarchy and readability.	One UI pass that normalizes button affordance, icon use, text density, empty states, and controller/keyboard focus.
5. Combat readability	Enemy telegraphs, impacts, shield states, beam heat, black holes, and asteroid danger need consistent readable language.	Audit every lethal or high-stakes action for visual cue, audio cue, HUD warning, and accessibility alternative.
6. Content surface cleanup	Some systems have one prototype item only: rare events, world events, missions, Forge registry, radar/shop progression.	Label prototype content clearly, then add two or three high-value variants only after the base presentation is coherent.

Systems, Menus, Objects, and Polish Matrix

System/Menu	State	Active objects/sprites/data	Needs polish	Suggestions
Run lifecycle and flow state	Implemented	Game flow states, run start, death, eject, continue, results, run result capture.	State transitions are broad and still largely orchestrated from GameScene. Edge-case language around completed mission plus continued exploration needs consistent treatment.	Add a run-state QA checklist: start, free range, mission complete, continue, eject cancel, eject confirm, death, restart, return to hub.
Splash and main command menu	MVP	Screen handles, cockpit backdrop, selected mission summary, run launch actions.	Needs a stronger first-read of current mode, selected ship, selected weapon, and why a run cannot start.	Make the command screen behave like a compact launch console: ship, contract/free-range, loadout, risk, launch readiness.
Pre-run hub navigation	MVP	Command, hangar, shop, settings tabs through preRunHubScreen.	Needs uniform navigation affordance, selected tab treatment, and back behavior across all pre-run screens.	Keep one persistent top/bottom nav component with visible current tab, disabled states, and consistent keyboard focus.
Ship select and hangar	MVP	Interceptor, Bulwark, Engineer; skins; stat ratings; starting weapons; loadout slots.	Engineer uses the Interceptor texture fallback. Ship fantasy/passives exist in copy but need stronger in-game proof. Mastery bonuses say coming soon.	Give every ship a unique Forge/cached sprite, one passive that visibly triggers, and a concise side-by-side stat comparison.
Loadout and weapon assignment	Implemented	Primary/secondary/auto slots, weapon inventory, starting weapon rules, hotbar assignment helpers.	Needs clearer slot identity and warnings when loadout rules block a selection.	Add icons for primary, secondary, auto, ship starter, locked, and source ship. Keep text short.
Shop and permanent upgrades	MVP	11 permanent upgrades, radar upgrades, sector scanner purchases, credits, active level controls.	Upgrade categories are useful but not yet emotionally interesting. Need clearer cost/value, maxed state, reset/active-level meaning, and progression balance.	Group shop into Ship, Weapon, Utility, Radar/Scanner. Show current effect, next effect, total owned, active level, and why an item matters.
Pause menu and settings	WIP	Pause, graphics, sound, controls, gameplay, accessibility tabs.	Several tabs are explicitly placeholder-heavy. Sound has no assets to control. Accessibility settings need real behavior.	Turn placeholders into either real toggles or hidden future entries. Prioritize screen shake, flash intensity, contrast telegraphs, text scale, volume, keybinds.
Controls and rebinding	MVP	Keyboard/mouse controls, binding slots, run control actions, awaiting binding state.	Needs controller support decision, focus handling for menus, and clear binding conflict feedback.	Add an input audit pass covering keyboard, mouse, gamepad plan, pause while binding, and reset defaults.
Debug menu and secret controls	Implemented	Tabs for run, player, ship, weapons, physics, collision, spawns, effects, black hole, visuals; secret overlay.	Developer surface is powerful but dense. Some labels expose implementation rather than testing intent.	Keep debug dense, but add scenario presets for common polish tests: low fuel, mission complete, heavy asteroid field, rare event, boss, shield build.
Test harnesses and diagnostics	Implemented	Smoke, phase harnesses 7 through 15.5, contact, beam, HUD mission log, result/fuel checks.	Harnesses are useful but scenario bodies still live deep in GameScene. Phase 16/17 visual acceptance is not represented as cleanly as gameplay harnesses.	Add visual smoke pages for HUD variants, Forge assets, radar levels, and menu screenshots. Extract harness scenario bodies later.

Results screen	MVP	Outcome title, rewards, build lines, upgrade lines, sector lines, continue/eject/death states.	Needs stronger learning value: why run ended, mission status, what rewards persisted, and suggested next action.	Add a clean report hierarchy: Outcome, Contract, Cargo, Upgrades, Damage/Threats, Sector, Unlocks.
Combat HUD/dashboard	WIP	GameplayHudSystem, fuel, hull, scrap, XP rail, mission button/log, weapons, scanner, radar, eject button, warning chips, variants.	Phase 17H has variants, but the final default needs one decisive pass for text fit, icon clarity, warning priority, and mobile/desktop framing.	Pick one variant, freeze it, then test at 1280x720, 1920x1080, and narrow/mobile ratios with screenshots.
Minimap and radar progression	WIP	MinimapSystem, radar levels 0-4, layer gating, signal markers, resources, threats, anomalies, scanner reveal.	Readable hierarchy has started, but full minimap redesign is deferred. Need player-facing explanation of radar levels.	Make radar levels unlock named layers with icons: Map, Objectives, Signals, Resources, Threats, Anomalies.
Sector scanner	MVP	Scanner runtime, scan timer, scanner targets, world/rare event targets, arrow graphics.	Needs clearer difference from radar, stronger scan feedback, and better target prioritization UI.	Use a timed pulse animation, scan completion sound, target category icon, distance/risk label, and shop upgrade preview.
Upgrade overlay	MVP	Three choices, reroll, banked upgrades, weapon summaries, keyboard selection.	Upgrade cards need exact before/after stat deltas, branch/evolution progress, rarity treatment, and clearer incompatibility/targeting.	For every card show affected weapon/system, current level, new level, exact stat change, and tags. Add icons via Forge.
In-run upgrade data	Implemented	Pulse, projectile, passive, beam, ramming upgrades; data-only Plasma Wake; many high-tier effects.	Most upgrade behavior is present, but balance/readability needs a pass. Plasma Wake is data-only and should be hidden or completed.	Create a build audit sheet: each upgrade has gameplay effect, visible effect, tooltip, audio cue, and balance note.
Progression and storage	MVP	Progression state, credits, unlock hooks, ship unlocks, weapon loadouts, radar/scanner levels, active permanent levels.	Needs fail-safe save migration, reset/export/import, and clearer distinction between owned unlocks and active power.	Add versioned save schema notes, in-game reset confirmation, and dev save snapshot export in debug menu.
Rewards and economy	Prototype	Scrap, credits conversion, reward resolver, mission/world/rare hooks, upgrade crates.	Rare parts are planned but not present. Scrap overlaps as XP, currency, and reward, so player meaning needs clarity.	Define resource roles: XP is in-run power, scrap is cargo/value, credits are post-run spend, rare parts are unlock milestones.
Free Range and missions	MVP	Free Range, Survey Signal, Salvage Cache, Enemy Probe, Mothership Contract, Rift Cache Contract.	Mission UI is functional but still contract-placeholder flavored. Mission completion needs more ceremony and map guidance.	Add mission briefing cards with objective, region, danger, reward, optional completion/end behavior, and radar/scanner promise.
Mission objective beacons	Prototype	Mission objective container, ring, core, reach-radius checks, objective labels.	Beacon needs final art, sound, distance/risk feedback, and different looks for survey/cache/probe/event contracts.	Create Forge mission beacon icons and world marker variants; add arrival pulse and completion state.
Sector generation	Implemented	2x default sector, scale options 1/2/3/5, regions, seed, safe drift, asteroid belts, salvage fields, enemy territory, anomaly signals.	Needs more visible regional identity and balancing of travel time, density, and fuel pressure.	Make each region visibly distinct in space and radar, then tune spawn/signal/resource density per region.

Sector runtime activation	Risk	Active sector asteroid/scrap/signal maps, weak maps, offscreen activation, coalescing, rollout timers.	Performance approach is good, but object lifecycle is still scene-owned and complex.	Keep behavior stable, but add metrics for active vs total sector objects and a debug overlay for activation radius.
World squads and encounter director	Implemented	Scout Pack, Gunner Escort, Strike Wing, Support Group, Sniper Screen, Carrier Group; activation/disengage.	Needs better world behavior identity: patrol, guard, ambush, reinforce, retreat, return to territory.	Add one clear behavior per squad type before adding more squads. Use radar labels and formation icons.
Unified enemy runtime	Implemented	15 lab enemy definitions, live enemies, behavior handlers, procedural visuals, Forge Scout asset.	Only Scout has a promoted Forge asset. Some behaviors need live-game balance/readability passes and audio cues.	Promote Forge art and telegraph/audio cue for each dangerous role in priority order: sniper, charger, exploder, support, carrier.
Legacy enemies	Risk	Basic, shooter, tank arrays and PNG sprites still exist beside live enemy runtime.	Compatibility is useful but increases bug risk when changing collision, minimap, rewards, and damage.	Decide whether legacy enemies are tutorial/fallback content or migrate fully to lab runtime.
World events	Prototype	Prototype Mothership, guard squads, hp, hit radius, danger radius, reward drops, contract guarantee.	Only one event exists. Mothership is stationary and not final boss art/balance. Stronghold kind is in type but no definition.	Add weak points, spawn bays, warning radius treatment, destruction stages, final art, and a separate Stronghold prototype.
Rare events	Prototype	Unstable Black-Hole Cache, Hunter Swarm, ambient spawn, guaranteed mission target, rewards.	Framework exists but event storytelling is light. Needs signal, danger, reward, and completion ceremony polish.	Make each rare event feel like a story: unique radar icon, intro ping, danger warning, local visual language, reward reveal.
Player flight and fuel	Implemented	Shared player flight runtime, velocity, thrust/strafe/brake, fuel drain only from thrust, emergency 10% thrust.	Needs final feel tuning per ship and clearer low/empty fuel feedback. Fuel pickups/caches are planned but absent.	Add low-fuel HUD/audio warning, emergency mobility tutorial cue, and later fuel caches as route decisions.
Player ships and skins	MVP	Interceptor implemented, Bulwark/Engineer MVP, skin tint data, unlock costs.	Ship art, passives, mastery, and cosmetic unlocks need final direction. Engineer needs unique runtime sprite.	Promote three core ship Forge assets, then add one mastery hook per ship and hide future mastery copy until real.
Pulse Cannon	Implemented	Held manual projectile weapon, primary/secondary support, Forge neon bolt, many upgrade effects.	Needs final sound, muzzle/impact polish, better upgrade visualization, and balance pass against squads/events.	Make Pulse Cannon the gold-standard weapon pass: muzzle flash, bolt trail, hit flash, upgrade deltas, audio, controller feel.
Ramming Shield	MVP	Shield image, dash charges, guard/bash damage, break effects, deflector/mirror/nova/aegis/stagger upgrades.	Needs better telegraph for shield HP, dash charges, bash window, blocked damage, and unsafe recovery.	Add shield arc health read, charge pips, impact sound, blocked-projectile sparks, and a short post-bash recovery cue.
Salvage Beam	MVP	Held beam, heat, overheat, tick damage, beam upgrades, tip sparks, ignition/vent effects.	Needs clearer heat UI, beam hit feedback, audio loop, and unique Engineer identity.	Add a beam heat meter in the weapon slot, color shift near overheat, audio ramp, and stronger contact sparks.

Projectile runtime	Implemented	Player, enemy, and world projectile types; pierce, range, statuses, black-hole capture, hit helpers.	Mostly solid, but future plasma trails need persistent area damage runtime. Enemy projectile readability needs audio/contrast.	Finish or hide Plasma Wake. Standardize projectile colors and silhouettes by owner/damage type.
Enemy projectiles	MVP	Shooter projectiles, live enemy projectile requests, sniper/bolt definitions, obstruction hits.	Danger level is not always readable before hit. Needs owner color, telegraph pairing, and shield interaction polish.	Give each enemy projectile class a shape, sound, color, hit spark, and optional warning line for high damage shots.
Physics, collision, world impacts	Implemented	Unified body helpers, world impacts, player contact runtime, spatial hash, hard/soft collision, knockback.	System is powerful but hard to reason about because callbacks and scene effects remain broad.	Keep tuning stable, add debug profiles for contact damage sources, and document each collision layer/mask contract.
Asteroids	Implemented	Tiers 1-10, four PNG variants, wrap mirrors, breakup recipes, coalescing, burst caps, damage flashes.	Gameplay is mature, but art/readability needs tier distinction and audio/impact clarity. PNG style differs from Forge direction.	Create Forge/cached asteroid tier language or retouched PNG set. Add tier-scaled cracks, chunk silhouettes, and impact sound.
Debris and wreckage	MVP	Enemy wreckage debris, destructible hazard behavior, death shards, throttled effects.	Normal enemy deaths no longer spawn persistent debris, but debris identity as hazard vs loot needs clearer visual language.	Use debris only for deliberate hazards and event fields. Give debris damage indicators and salvage color separate from scrap.
Scrap pickups and loot	Implemented	Scrap tiers 1-4, tier rollup, scrap as resource and XP source, upgrade crates.	Needs clearer pickup values, magnet/vacuum feedback, and differentiation between scrap, credits, XP, and upgrade crates.	Add pickup value burst labels, resource icons, and a short HUD cargo pulse when scrap converts to rewards.
Black holes and world forces	MVP	BlackHoleSystem, gravity/whirlpool, capture, consumption, warning visuals, debug tuning, field saves.	Needs final visual identity, audio warning, performance guardrails, and consistent interaction with enemies/projectiles/debris.	Keep cached/simple visuals. Add distance warning, pull line language, low-frequency audio, and rare-event-specific treatment.
Combat feedback and effects	MVP	Floating damage, impact explosions, shield feedback, beam sparks, asteroid ghost/grow-in, death shards.	Needs hierarchy so important feedback beats decorative feedback. Accessibility options are not wired enough.	Classify effects as critical, combat, reward, ambient. Add density scaling and high-contrast mode.
Starfield/background/camera	Implemented	StarfieldSystem, parallax layers, camera lead, wrap-aware rendering.	Background is functional but could better communicate region identity and speed without clutter.	Add subtle region tint/debris/signal motifs and keep parallax performance measured.
Asset Forge pipeline	WIP	Forge schema, renderer, SVG/export/import, AI task protocol, editor, promotion bundles, five implemented registry assets.	Pipeline is promising but only initial acceptance assets are promoted. In-app provider calls and direct source patching are deferred.	Use Forge as the required path for all new promoted ships, enemies, projectiles, pickups, UI icons, mission icons, and radar icons.
AI candidate asset folders	WIP	Generated candidates for HUD frames, warning chips, weapon slot frames, resource/stat/weapon icons, ships, enemy ships, bosses.	Candidates are not runtime assets. They need curation, background cleanup, slicing, scale tests, and promotion decisions.	Create a candidate review board: Accept, Revise, Reject, Promote to Forge recipe, Promote to PNG fallback.

Audio	Missing/Mentioned	No mp3/wav/ogg/flac/m4a assets found in the workspace.	All weapon, impact, UI, warning, pickup, mission, enemy, black-hole, and result sounds are absent.	Start with a tiny reusable sound set and settings sliders: master, music, SFX, UI. Then add per-enemy telegraph cues.
Accessibility	Missing/Mentioned	Pause/settings accessibility placeholders; planned reduced shake, high contrast, flash/text scale, input assist.	Needs real toggles tied to rendering and gameplay feedback, not just menu labels.	Implement reduced flash, reduced shake, high-contrast telegraphs, text scale, color-safe enemy shots, and optional auto-fire/aim assist.
Desktop bridge and packaging	MVP	Electron scripts, desktop bridge, desktop build script, dev desktop command.	Steam/desktop release polish is post-prototype. Needs full-screen/window options, save location behavior, icon, installer, crash logging.	Keep browser as primary until loop polish is stronger. Later run desktop smoke separately from web build.
Performance profiling	Implemented	PerformanceProfilerSystem, report export, counts/flags, previous lag report, debug diagnostics.	Needs a routine benchmark suite before adding more visuals and audio.	Add canned 2-minute stress runs: asteroid burst, swarm, mothership, rare event, high-upgrade Pulse, beam, black hole.

Active Runtime Object and Sprite Inventory

Scene-owned active objects

Object group	Current active objects	Polish note
Player	Player container, player image, player velocity, ramming shield image, thruster timing, hull, invulnerability, fuel.	Needs unique ship sprites, clearer thruster/fuel feedback, and per-ship feel polish.
Weapons/projectiles	Player projectiles, enemy projectiles, beam slot graphics/state, muzzle/trail/impact effects, black-hole captured projectiles.	Needs final owner color language, audio, and upgrade-specific visual feedback.
Enemies	Legacy basic/shooter/tank arrays, live lab enemy array, enemy bodies, wrap mirrors, behavior state, squad membership.	Needs single final runtime ownership strategy and full Forge art coverage.
World hazards	Basic asteroids, asteroid mirrors, debris containers, death shards, black hole system.	Needs visual cohesion and danger readability across asteroid tiers, debris, and black holes.
Pickups	Scrap pickups, active sector scrap, upgrade crate pickup texture/key, pickup feedback.	Needs icon/value clarity and future fuel/rare-part pickup identity.
Sector content	Sector layout, asteroid/scrap/signal spawns, active sector maps, signal beacons, mission objective beacon.	Needs clearer region and signal art, plus more obvious discovered/undiscovered state.
Events	World events, rare events, mothership visuals, rare-event completion/reward feedback, guard squads.	Needs final event spectacle, warning language, and event-specific sprites/icons.
UI/HUD	Gameplay HUD, minimap, sector scanner arrow, upgrade overlay, results button/screen, pause menu, shop, ship select, debug menu, secret overlay.	Needs final visual hierarchy, icon replacement, consistent focus/click states, and settings that do real work.

Current asset files and runtime texture families

Asset family	Found assets	State	Polish direction
Player ships	spaceship_1, spaceship_2, spaceship_3, bulwark, bulwark2, bulwark_teeth; Forge Interceptor.	Mixed	Promote Interceptor, Bulwark, Engineer as cohesive Forge/cached assets. Decide whether old PNGs are fallback or retired.
Enemy ships	enemy_chaser, enemy_shooter, enemy_tank, spaceship_enemy_1; procedural lab visuals; Forge Scout.	Mixed	Promote Forge art for all 15 live enemy roles and keep procedural fallback only for lab editing.
Asteroids	astroid_1 through astroid_4 PNGs; tier system is gameplay-driven.	Implemented	Add tier-distinct silhouettes/marks and consistent damage/breakup visual language.
Debris and scrap	deabri, scrap, scrap_tier_1/2/3/4 PNG and SVG tier assets; Forge tier-1 scrap.	Implemented	Forge or polish all scrap tiers and make debris visually separate from collectible scrap.
Weapons	ramming shield PNG; Pulse projectile Forge neon bolt; beam is code-rendered.	MVP	Add weapon icons, stronger beam/shield art, muzzle/impact sprites, and audio identity.
Upgrade UI	upgrade_create PNG; upgrade overlay drawn by Phaser UI.	MVP	Create upgrade icons by category/weapon/rarity. Use exact stat deltas.
HUD/UI candidates	AI candidates for status readout, warning chip, weapon slot frame, resource/stat/weapon icons.	WIP	Review and promote a selected set. Avoid baking unreadable text or white backgrounds into runtime UI.
Audio	No audio files found.	Missing	Create a minimal sound set before deeper combat polish.

Missing or Mentioned Objects, Sprites, Menus, and Systems

Missing/mentioned item	Where it fits	Current implication	Suggested treatment
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Stronghold event	World events and missions	Type exists as a world event kind, but only mothership-prototype is defined.	Add a first stronghold with turrets/bays/weakpoints after mothership polish.
Moving mothership, spawn bays, weak points, final boss art	World events	Prototype mothership is stationary and structurally simple.	Phase it: warning radius, guard waves, weakpoints, bay spawns, movement, final art.
Rare events: Thousand Swarm, Dead Fleet, Command Net, Reactor Cascade, Sniper Constellation, Mirror Shield Convoy	Rare event/content expansion	Named or implied by the 2.0 plan but not implemented.	Add only after Unstable Black-Hole Cache and Hunter Swarm feel special.
Additional mission templates	Mission system	Destroy Target, Recover Object, Survive Zone, Clear Stronghold, Hunt Rare Enemy, Investigate Signal, Assassinate Commander, Escape Pursuit, Destroy Mothership are planned categories; current set covers only a subset.	Build one mission per proven event/region type, not many shallow contracts.
Future ships: Carrier, Void Skiff, more unlockable variants	Ships/loadouts/progression	Only Interceptor, Bulwark, Engineer are active.	Do not add until the three core ships have unique art, passives, and balance.
Ship mastery bonuses	Ship progression	All three ships mention level 20 mastery coming soon.	Hide or implement mastery preview. Avoid promising invisible progression.
Fuel cells, fuel caches, fuel upgrades/modules	Fuel/economy/world resources	Fuel is thrust-use-only; world fuel drops are planned but absent.	Add rare fuel caches after low-fuel feedback is readable.
Rare parts	Economy/unlocks	Planned as a resource but not present in current data flow.	Use rare parts only for meaningful unlocks, not stat grind.
Controller support	Controls/accessibility	Planned historically, not required for first playable prototype.	Design focus/firing/aim behavior before wiring controller UI.
Audio categories and assets	Phase 18	No audio assets currently exist.	Build sound categories and sliders together with first sound pack.
High-contrast telegraphs, reduced shake/flash, aim assist/auto-fire	Accessibility	Planned in Phase 18 and partially represented as settings placeholders.	Implement as real gameplay/render toggles and add harness/screenshot coverage.
Forge assets for all non-Scout enemies	Asset pipeline/enemies	Only Scout has a promoted Forge asset.	Prioritize Sniper, Wedge Striker, Reactor Drone, Shield Frigate, Carrier Foundry.
Forge assets for Bulwark and Engineer	Ships/assets	Interceptor has Forge asset; Bulwark uses PNG; Engineer uses fallback ship texture.	Promote unique ship assets before adding new ships.
Mission, radar, resource, stat, weapon, upgrade, telegraph icons	UI/assets	Some AI candidate sheets exist, but runtime icon coverage is incomplete.	Create a locked icon set and use it consistently in HUD, shop, missions, and upgrade overlay.
Plasma Wake runtime	Pulse upgrade/effects	Data-only upgrade notes say persistent area-damage trail runtime is required.	Either implement area trail with performance cap or keep hidden from upgrade pool.
Full data-only off-screen pickup storage	Scrap/performance	Deferred from scrap rollup work.	Only revisit if active pickup counts become a problem after visual polish.
Direct in-app AI provider calls	Asset Forge	Manual export/import exists; provider calls are deferred.	Keep manual flow until promotion quality standards are stable.
Direct browser-side source patching for asset promotions	Asset Forge/dev tooling	Promotion bundles are exported but source patching is manual.	Keep manual patching unless promotion volume becomes high.

Recommended Polish Roadmap

Pass	Scope	Definition of done
Pass A: Phase 17 closeout	HUD, radar, mission log, results, menu consistency, screenshots.	One default HUD variant, readable mission/free-range states, results explain outcome and rewards, build passes, visual smoke captured.
Pass B: Core asset cohesion	Forge assets for three ships, 15 enemy roles, scrap tiers, weapon/mission/radar/upgrade icons.	Every common active object has either promoted Forge art or an intentional fallback label.
Pass C: Phase 18 audio/feedback/accessibility	Sound categories, first sound pack, high-contrast telegraphs, reduced shake/flash, low-fuel and enemy danger cues.	Every lethal or high-stakes action has visual and audio feedback plus an accessibility-friendly alternative.
Pass D: Content prototype uplift	Mothership, rare events, missions, scanner/radar rewards, shop clarity.	Each prototype system has at least two polished examples or is clearly marked as future.
Pass E: Balance and performance sweep	Ships, weapons, upgrades, enemies, asteroids, events, economy, sector density.	Stress harnesses pass, no major frame spikes, and each ship/weapon has a distinct viable path.

Immediate Suggested Task List

1. Freeze the Phase 17 HUD/dashboard variant and remove or demote research variants from normal player paths.
2. Create a screenshot smoke set for command, hangar, shop, pause/settings, upgrade overlay, HUD, minimap/radar levels, results, and Enemy Lab Forge.
3. Promote unique Forge/cached sprites for Bulwark and Engineer, then replace Engineer's Interceptor texture fallback.
4. Promote Forge enemy assets for Wedge Striker, Needle Sniper, Reactor Drone, Shield Frigate, and Carrier Foundry.
5. Build the first audio pass and wire real sound sliders.
6. Turn accessibility placeholders into real settings: reduced shake, reduced flash, high-contrast telegraphs, text scale.
7. Add mission briefing polish and mission completion ceremony for reach-location, world event, and rare event objectives.
8. Give the mothership weakpoints or spawn bays before adding another world anchor.
9. Hide or finish data-only Plasma Wake and "coming soon" mastery promises.
10. Run a stress/build pass after each visual/audio batch to protect the current performance gains.